

TRIMBLE

Trimble is a global product and technology company specializing in construction, transportation, navigation, and geospatial solutions. Its offerings like Trimble Maps provide industrial-grade routing, mapping, and fleet navigation, integrating physical site data with digital analytics to help industries plan, measure, and manage projects more efficiently. Its tools are used in infrastructure projects including Chennai Metro Rail for rail asset management, GPS & BIM-based construction, and fleet operations.

Role: Software Engineer

Interview Process:

- Online Assessment
 - Technical Interview
 - Managerial Round
 - HR round
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Online Assessment:

The online test was conducted on the Codility platform, where we had two coding questions.

Question 1: You are given a 2D matrix of integers. From this matrix, you can select any rectangular submatrix (a block formed by continuous rows and columns). The task is to count how many such rectangles exist where the sum of the numbers inside the rectangle is exactly equal to the sum of the numbers outside the rectangle. In other words, the selected rectangle should split the matrix into two parts with equal sums - the inside and the outside.

Question 2: You are given a string that may contain lowercase letters and the special character '?', along with an integer K. The task is to replace all the '?' characters with lowercase letters and, if needed, change at most K other characters in the string so that the final result becomes a palindrome. If it is possible to form such a palindrome under the given conditions, return the resulting string; otherwise, return "NO".

Technical Interview:

- The interview lasted for about 1 hour and 15 minutes. It started with a short self-introduction, one of my project explanation and after which the interviewer moved to OOPs concepts. I was asked to explain the four pillars of OOPs along with real-world examples. This was followed by a rapid-fire round on other OOPs concepts.
- I was then asked to write code in my programming language to demonstrate method overloading and method overriding.
- Next, the interviewer switched to problem-solving. The first task was to write a program to find the highest occurring element in an array along with its frequency, but with a strict constraint - I was not allowed to use any inbuilt data structures (like maps, vectors) or functions. I had to solve it using only a single array and a loop.
- In the data structures section, I was given a question on binary trees: to find the minimum depth of a binary tree. After that, I was asked to give real-world examples of where data structures like linked list, stack, queue, graphs, and trees are used.

- The discussion then moved to dbms, where I was asked to write queries for operations such as create, insert, alter, update, and modify on a given sql table, followed by questions about my overall knowledge in sql.
- To cover the computer networks domain, the interviewer presented a real-world banking example and asked me to identify the OSI layer used in that case. In Operating Systems, the questions were more general, such as the difference between process and thread, what I know about synchronization, and deadlocks.
- Towards the end, an aptitude question was given: There are coins of 5, 2, and 1 rupees. Find the minimal number of coins needed to make exactly 100 rupees.
- The interview was very interactive and engaging throughout. I was able to answer almost all the questions confidently, except for the banking scenario in networks. The interviewer was supportive and guided me whenever needed, which made the overall experience very positive.

After about 10 minutes, I received the news that I was selected for the Managerial Round.

Managerial round:

The interview lasted for about an hour. It began with a self-introduction, and right from the start, the interviewer was very polite and humble, which gave me the confidence to speak more freely.

He asked me to share about my school life, followed by why I choose ceg-cse and my interest in joining Trimble. He also inquired about my study methods, preparation strategies for interviews, and gradually made it clear that this round was mainly to test my communication, attitude, and confidence rather than technical depth.

The interviewer also asked me about my native place and family background. At one point, he gave me a small coding snippet task – to find the berth position in a train – while simultaneously asking me casual questions about my college life.

Later, he moved on to technical concepts, asking me the difference between a Binary Tree and a Binary Search Tree, and asked me to take examples and explain searching in both structures.

After that, he asked me to explain all the projects I had mentioned in my resume and to go into detail about one specific project. He carefully went through my resume, asking questions about my extracurricular activities, the clubs and organizations I was part of, and my roles and responsibilities in them.

As a whole, it was a very positive and comfortable experience where I felt free to express my thoughts and knowledge.

About ten minutes later, I was informed that I was selected for the HR Round.

HR round:

The HR round lasted for about 30-45 minutes. It started with a self-introduction, followed by questions on what I know about Trimble and why I wanted to join the company.

The interviewer then asked me to explain my Deep Learning project in detail and discussed my interest in AI/ML/DL. Along with this, I was also asked about my family background and native place.

Next, the interviewer focused on my overall journey in the process from the coding test to the technical and managerial rounds, and how I felt about the entire experience. A behavioral question was also asked to understand my approach and mindset.

The round was interactive and comfortable, allowing me to express myself freely.

In the end, out of 11 candidates who attended the HR round, 6 were selected, and I was glad to be one among them.

Company specific preparation tips:

During the process, I realized that it is important to listen carefully to the PPT, as some of the questions can be asked from it during hr round. In the coding round, which was held on the Codility platform, an AI assistant was provided to test our prompting skills. The idea was that if you decide to use it, you should know how to give the correct prompts; otherwise, it is equally fine to solve without it, as I did. Though the overall process might seem a bit lengthy, having an intermediate level of knowledge across all core computer science subjects along with decent practice in data structures and algorithms is more than enough to do well. The Managerial round may sometimes involve system design questions, even though I was not asked any, so it is advisable to go through at least the basic concepts of system design in advance. Overall, the key lies in being confident, clear with your concepts, and expressive in communication throughout the interview journey.

Personal note:

This was actually my second interview and honestly through the whole preparation phase I understood one thing - consistency matters a lot. Whatever you prepare, just keep revising it again and again. Be clear in your answers and at the same time try to improve your communication skills along the way. Don't feel bad if something doesn't go well, and most importantly, don't stop learning. Even clearing the online test itself is not that easy, so once you get the chance to sit for an interview, just grab that opportunity . All the best !

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